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AI Recruitment Screening- System

BUSINESS ANALYSIS & PROJECT MANAGEMENT CASE STUDY

1. Executive Summary

This report presents a Business Analysis and Product Management case study for an **AI Recruitment Screening System** designed to improve the efficiency, consistency, and quality of early-stage hiring. The proposed solution acts as an AI-assisted decision-support layer for recruiters by automating CV parsing, candidate-job matching, candidate scoring, and ranked shortlisting while preserving human oversight for final hiring decisions.

The case study explores the business problems, stakeholder needs, future-state recruitment process, product scope, requirements, AI system design, validation approach, and risk and ethics considerations. It is positioned as a practical recruitment optimization solution for organizations handling medium to high volumes of applicants, especially in technology and knowledge-based hiring contexts.

Rather than replacing recruiters, the proposed system is intended to **augment recruitment workflows** by reducing repetitive manual screening effort and supporting more structured shortlisting decisions. This report consolidates the key business, product, and system design outputs produced throughout the case study.

2. Project Background

Recruitment is one of the most resource-intensive functions in many organizations, particularly during the early stages of hiring where recruiters must review large volumes of applications within limited timeframes. In traditional hiring environments, CV screening is often performed manually by recruiters who compare candidate resumes against job descriptions, identify relevant qualifications, and shortlist applicants for the next stage of the hiring process.

This approach becomes increasingly difficult to sustain when application volumes rise. Manual screening is time-consuming, repetitive, and highly dependent on individual recruiter judgment. As a result, organizations may experience slow time-to-hire, inconsistent evaluation standards, and increase risk of overlooking qualified candidates. These challenges are especially significant in high-demand sectors such as IT, where large applicant pools and role-specific skill requirements make screening more complex.

The purpose of this case study is to explore how Artificial Intelligence can be applied responsibly to improve the recruitment screening stage. The project focuses on designing an AI Recruitment Screening System that can assist recruiters by automatically processing candidate CVs, comparing them against job requirements, and generating ranked recommendations while still preserving human decision-making authority.

3. Problem Statement

Organizations face inefficiencies in manual recruitment screening due to high application volumes, limited recruiter capacity, inconsistent evaluation methods, and the risk of unconscious bias in early-stage shortlisting. Recruiters are often required to manually review each candidate's CV, interpret job relevance based on experience and skills, and make quick shortlisting decisions under time pressure.

This process creates several business challenges:

- Long recruitment cycles due to the time required for manual CV review
- Inconsistent candidate evaluation caused by subjective interpretation across recruiters
- High recruiter workload driven by repetitive screening tasks
- Scalability limitations when hiring for multiple roles or processing large numbers of applications
- Risk of overlooking qualified candidates due to fatigue, time pressure, or inconsistent screening criteria
- Potential for bias in decision-making when evaluation is not standardized

As organizations increasingly seek faster and more data-driven hiring processes, there is a need for a scalable and structured screening solution that can automate repetitive recruitment tasks, evaluate candidates more consistently, and support recruiters in making more informed shortlisting decisions.

4. Project Objectives

The primary objective of this case study is to design an AI-powered Recruitment Screening System that improves the speed, consistency, and quality of candidate shortlisting while keeping human recruiters and hiring managers in control of final decisions.

The specific objectives of the project are:

- To reduce manual CV screening effort by automating the extraction and evaluation of candidate information
- To improve recruitment efficiency by generating ranked candidate recommendations based on job relevance
- To standardize early-stage candidate evaluation through structured and repeatable matching criteria
- To support fairer and more transparent screening by reducing dependence on purely subjective first-pass reviews
- To provide recruiters with a dashboard for reviewing AI-generated results, filtering candidates, and manually overriding recommendations where necessary
- To design a solution concept that is scalable and suitable for organizations handling medium to high volumes of recruitment activity
- To demonstrate how Business Analysis, Product Management, and AI system design can be integrated within a single end-to-end case study

These objectives ensure that the proposed solution addresses both operational recruitment challenges and broader product and business value considerations.

5. Scope Definition

The scope of this case study focuses on the design of an AI-assisted recruitment screening solution rather than a full end-to-end HR or hiring platform. The goal is to define a realistic product scope that addresses the early screening stage of recruitment while remaining aligned with a Business Analysis and Product Management perspective.

In Scope

The following capabilities are included within the scope of the proposed AI Recruitment Screening System:

- Candidate CV upload and storage
- CV parsing and extraction of structured candidate information such as skills, education, certifications, and work experience
- Job description submission and processing to extract required skills, qualifications, and experience criteria
- Candidate-job matching using structured evaluation logic and AI-assisted relevance scoring
- Candidate ranking based on calculated job fit scores
- Recruiter dashboard for viewing ranked candidates and screening results
- Filtering, sorting, and shortlist support for recruiters
- Recruiter override functionality to ensure human review remains part of the decision process
- High-level reporting of candidate screening results and shortlisting outputs

Out of Scope

The following areas are intentionally excluded from the scope of this case study:

- Interview scheduling and calendar management
- Video interview analysis or interview transcription features
- Payroll, onboarding, or broader HR management functionality
- Background verification and reference checking
- Offer management and employment contract workflows
- Full replacement of existing Applicant Tracking Systems (ATS)
- Development of a production-grade machine learning model with real training data

By defining clear boundaries, the project remains focused on solving the specific business problem of inefficient early-stage recruitment screening.

6. Stakeholder Analysis Summary

The success of the AI Recruitment Screening System depends on understanding the needs, expectations, and concerns of the stakeholders involved in the recruitment process. The stakeholder analysis identified both direct users of the system and supporting stakeholders who influence the product's adoption, governance, and long-term effectiveness.

Primary Stakeholders

HR Managers: HR Managers are responsible for recruitment strategy, process performance, and hiring efficiency across the organization. Their interest in the system lies in reducing time-to-hire, improving recruiter productivity, and ensuring that recruitment operations remain scalable and aligned with organizational hiring goals.

Recruiters: Recruiters are the primary operational users of the system. They interact directly with the screening workflow by uploading or reviewing job descriptions, examining ranked candidates, filtering results, and making shortlisting decisions. Their needs include speed, usability, transparency of AI recommendations, and the ability to override system-generated results when necessary.

Hiring Managers: Hiring Managers define role-specific requirements and are responsible for approving interview-ready candidates. They rely on recruiters to provide strong shortlists and therefore benefit from a system that improves the quality and consistency of candidate selection. Their key interest is ensuring that the AI system identifies candidates who are genuinely aligned with the job requirements.

Candidates: Candidates are external stakeholders whose applications are processed by the system. Although they do not directly interact with the internal dashboard, they are directly affected by the fairness, consistency, and accuracy of screening outcomes. Candidate interests include equal opportunity, transparent evaluation, and responsible handling of personal data.

7. **Business Process Analysis**

The business process analysis examined both the **current-state (AS-IS)** recruitment workflow and the **future-state (TO-BE)** workflow supported by the proposed AI Recruitment Screening System.

AS-IS Process

In the current-state recruitment process, recruiters manually review candidate CVs after a job is posted and applications are received. Each CV must be read, interpreted, and compared against the job description to determine whether the candidate should be shortlisted. This process is heavily dependent on recruiter's time and judgment.

The AS-IS process generally follows these steps:

1. Job vacancy is created and published
2. Candidates submit applications and CVs
3. Recruiters manually review each CV
4. Candidate qualifications are compared against job requirements
5. Recruiter decides whether to shortlist or reject the application
6. Shortlisted candidates move to the next recruitment stage

Key Problems in the AS-IS Process

- Screening is slow and repetitive
- Candidate evaluation may vary across recruiters
- Large application volumes create bottlenecks
- Strong candidates may be overlooked due to time pressure or inconsistency
- Recruiter workload increases significantly during high-volume hiring periods

TO-BE Process

The future-state process introduces AI as a **decision-support layer** in the screening workflow. Instead of manually reading every CV from the start, recruiters can use the system to automatically structure candidate data, compare candidates against job requirements, and generate ranked candidate recommendations.

The TO-BE process follows this flow:

1. Recruiter submits or uploads a job description
2. Candidates upload CVs through the recruitment channel
3. The system parses CVs and extracts structured candidate information
4. The system processes the job description and identifies key hiring criteria
5. The AI matching engine evaluates candidate-job relevance and assigns scores
6. Candidates are ranked based on job fit
7. Recruiter reviews ranked results, apply filters, and manually confirms or overrides shortlisting decisions

Business Process Improvement

The TO-BE process improves recruitment operations by:

- Reducing time spent on manual first-pass screening
- Introducing more consistent evaluation logic
- Allowing recruiters to focus on review and decision-making rather than repetitive reading
- Supporting scalability for high-volume hiring
- Maintaining human control over final shortlist decisions

This process analysis demonstrates how the proposed solution directly addresses the inefficiencies identified in the current recruitment workflow.

8. Solution Overview

The proposed AI Recruitment Screening System is designed as an AI-assisted recruitment decision-support platform that improves the efficiency and consistency of candidate shortlisting. It is intended to support recruiters rather than replace them, automating repetitive screening activities while preserving human oversight over hiring decisions.

The solution is centered around the idea that candidate CVs and job descriptions can be transformed into structured, comparable data. Once both candidate and job information are standardized, the system can evaluate job relevance, generate candidate scores, and present ranked results to recruiters in a way that supports faster and more informed shortlisting.

Core Solution Components

1. CV Upload and Parsing

Candidates or recruiters upload CV documents into the system. The system extracts key information such as:

- Skills
- Education
- Work experience
- Certifications
- Relevant role history

This converts unstructured resume content into a structured candidate profile.

2. Job Description Processing

Recruiters submit or upload a job description. The system extracts:

- Required skills
- Experienced expectations
- Educational qualifications
- Domain-specific keywords
- Optional or preferred attributes

This creates a structured representation of the job's screening criteria.

3. Candidate-Job Matching

The matching engine compares the structured candidate profile against the structured job requirements using weighted logic. The objective is to estimate how relevant each candidate is for the role.

4. Candidate Scoring and Ranking

Each candidate is assigned a score based on their fit with the role. Candidates are then ranked from highest to lowest relevance.

5. Recruiter Dashboard

Recruiters view the ranked results through a dashboard that allows them to:

- Review candidate scores and ranking positions
- Filter and sort results
- Identify top candidates quickly
- Manually override or confirm shortlist decisions

Solution Positioning

The solution is positioned as a decision-support tool, not an autonomous hiring engine. Human recruiters remain accountable for the final decision, while the system accelerates the early screening stage and improves consistency in how candidates are evaluated.

9. Requirements Summary

The AI Recruitment Screening System requirements were documented through a BRD, SRS, functional requirements, non-functional requirements, user stories, and acceptance criteria.

The system is expected to:

- Allow CV uploads and job description submission
- Parse candidate and job data into structured information
- Match candidates against job requirements
- Generate candidate scores and rankings
- Provide recruiters with a dashboard to review and filter results
- Allow recruiters to override AI-generated recommendations

Non-functional requirements focus on performance, scalability, security, usability, and reliability, ensuring the system is practical for real recruitment environments.

10. Product Management Approach

From a Product Management perspective, the system was treated as an AI-enabled recruitment product rather than just a technical concept.

The product vision is to help organizations hire faster, smarter, and more consistently through AI-assisted screening. The MVP focuses on the core screening workflow:

- CV parsing
- Job description processing
- Candidate matching
- Candidate ranking
- Recruiter dashboard

A phased roadmap was also defined, beginning with MVP delivery and extending toward future improvements such as enhanced matching, analytics, and integration capabilities.

11. AI System Design Summary

The AI Recruitment Screening System was designed using a layered architecture consisting of:

- **Presentation Layer** – recruiter dashboard and user interface
- **Application Layer** – CV processing, job processing, and matching services
- **AI Layer** – candidate-job matching and scoring logic
- **Data Layer** – candidate, job, and scoring data storage

The AI matching process compares candidate qualifications with job requirements based on skills, experience, education, and relevance. The output is a ranked list of candidates presented to recruiters for review.

The system is designed as a **decision-support tool**, meaning AI assists screening but does not replace human hiring decisions.

12. Testing and Validation Approach

The testing and validation approach focuses on ensuring that the system is functionally correct, useful to recruiters, and aligned with business expectations.

User Acceptance Testing (UAT) scenarios were created for:

- CV upload and parsing
- Job description submission
- Candidate scoring and ranking
- Dashboard review
- Recruiter override functionality

Validation also considers whether the system improves screening efficiency, produces relevant candidate rankings, and remains easy for recruiters to use.

13. Risk, Compliance and Ethics

Since the system operates in a hiring context, risk and ethics are important considerations.

Key risks include:

- Incorrect candidate ranking
- Parsing errors from inconsistent CV formats
- Recruiters over-relying on AI recommendations
- Data privacy and security concerns

To address these risks, the system is designed with:

- Human oversight and recruiter override capability
- Focus on job-relevant evaluation criteria
- Secure handling of candidate data
- Transparency in AI-assisted decision support

The solution is positioned as a **responsible recruitment support system**, not an autonomous hiring decision-maker.

14. Business Value and Expected Impact

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15. Conclusions

The proposed **AI Recruitment Screening System** presents a practical and forward-looking solution to one of the most time-consuming challenges in modern recruitment: early-stage candidate screening. By combining **Business Analysis, Product Management, and AI system design**, this case study demonstrates how recruitment workflows can be improved through structured automation while still preserving the critical role of human judgment.

The system is designed to address key recruitment pain points such as high application volumes, slow manual screening, inconsistent evaluation, and limited scalability. Through capabilities such as **CV parsing, job requirement extraction, candidate-job matching, scoring, and ranking**, the solution enables recruiters to identify relevant candidates more efficiently and make more informed shortlisting decisions. At the same time, the inclusion of **human oversight, recruiter override functionality, and ethical safeguards** ensures that the system remains a decision-support tool rather than an autonomous hiring mechanism.

Overall, the AI Recruitment Screening System delivers value not only as a technology concept, but as a well-structured business solution that aligns with operational recruitment needs, product strategy, and responsible AI use. It demonstrates how AI can be applied in a meaningful way to enhance recruiter productivity, improve consistency in candidate evaluation, and support more scalable hiring processes. As a case study, it reflects a complete end-to-end approach to designing an AI-enabled product with both business impact and practical implementation value.